SUNNY GEORGE CHANDY

DIRECTOR, AI/ML ENGINEERING | ENTERPRISE AI & ENGINEERING LEADERSHIP

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EXECUTIVE PROFILE

AI engineering leader with 15+ years of experience building global engineering organizations and shipping enterprise software and production AI across regulated healthcare and financial services. Lead a 40-member organization spanning AI engineering, application development, quality engineering, SRE, and operations. Own AI strategy, architecture, roadmaps, delivery, governance, and production reliability. Combine hands-on agentic AI, RAG, and evaluation architecture with engineering modernization, accountable team ownership, and measurable business outcomes.

LEADERSHIP FOCUS

Global Engineering Leadership • AI Strategy & Architecture • Agentic AI & LLM Evaluation • Software Delivery • Cloud & SRE • DevOps & Observability • Responsible AI

PROFESSIONAL EXPERIENCE

Director, AI/ML Engineering

Optum Financial Services (Optum) | Minneapolis, MN | Jul 2024 – Present

Lead a 40-member global engineering organization delivering conversational and agentic AI, software engineering, QE, SRE, and production operations across Azure and Google Cloud.

- **Production AI delivery.** Delivered conversational AI supporting 220K+ daily interactions and 3M+ monthly conversations with 75% containment, 94% customer satisfaction, 99.99% uptime, and $8M+ in reported annual operational savings.
- **AI strategy & organizational leadership.** Define enterprise AI architecture, engineering priorities, ownership standards, production-readiness practices, and delivery roadmaps; partner with Product, Operations, Security, and executives to prioritize business outcomes.
- **Agentic AI architecture.** Architected and led an omnichannel multi-agent employer platform using Google ADK, routing/retrieval/task agents, short- and long-term memory, model routing, MCP integrations, and grounded retrieval [OKF, vector database]. Improved IVR containment from 50% to 65% and generated $385,000 in call cost avoidance.
- **Docusage.** Built an enterprise AI document intelligence platform for Optum Financial Services that increased straight-through processing from 50% to 80%, with approximately $1 million in projected annual operating savings.
- **AI evaluation & responsible deployment.** Built automated LLM-as-judge, rules-based and human-review evaluation using golden datasets and labeled conversations; measured groundedness, accuracy, relevance, hallucinations, safety, and tool-use success.
- **Governance & adoption.** Partner with security, compliance, and architecture teams on review-board standards, production controls, and deployment readiness; building a Forward Deployed Engineering practice to speed use-case discovery and adoption.
- **SRE, infrastructure & developer platforms.** Lead the SRE team accountable for cloud infrastructure provisioning, CI/CD pipeline engineering, deployment automation, observability, service reliability, and operational readiness for live AI applications.
- **AI observability.** Used Datadog and Arize to monitor and evaluate agent performance.
- **Engineering quality & performance.** Built an AI-persona performance-testing platform using k6 to simulate 100–1,000 concurrent users and independently evaluate response quality, latency, scalability, and reliability.
- **Customer experience intelligence.** Delivered conversational analytics including transcription, resolution detection, repeat-caller patterns and complaint categorization, helping operational teams identify friction and improve service quality.

PROFESSIONAL EXPERIENCE — CONTINUED

Senior Engineering Manager – Software Engineering & Operations

Optum Pay (Optum) | Minneapolis, MN | Apr 2017 – Jun 2024

Led software engineering, QE, DevOps, SRE, platform operations, and modernization for healthcare payment services processing $300B+ annually for 2M+ healthcare professionals.

- **Engineering operations.** Led multidisciplinary teams across development, performance engineering, releases, and 24×7 production support; sustained 99.995% availability through resilient engineering and operational controls.
- **End-to-end observability.** Implemented application-wide end-to-end observability, strengthening visibility across services and infrastructure to support proactive monitoring, incident investigation, and production reliability.
- **Quality engineering & delivery.** Led development, QE, SRE and IT operations together, with accountability across application lifecycle, test automation, release management, incident response and continuous improvement.
- **Cloud & infrastructure modernization.** Led enterprise Azure migration with zero SLA breaches and established Optum Financials' first chaos engineering practice to surface failure modes and strengthen resilience.
- **Security transformation.** Reduced critical/high vulnerabilities by >99% (from 100K+ to fewer than 100), combining engineering prioritization, remediation, and controls.
- **Performance & resiliency.** Directed performance testing, production capacity planning, and reliability improvement for distributed healthcare payment platforms serving more than 2M providers.
- **Critical healthcare infrastructure.** Scaled infrastructure supporting $135.2B in HHS COVID-19 Provider Relief Fund payments to 421K+ healthcare providers.
- **Crisis analytics & reporting.** Delivered dashboards tracking $6.5B in Temporary Funding Assistance during the Change Healthcare outage, improving operational visibility into relief payments.

IT Analyst – Software Engineering

Tata Consultancy Services (Client: Target) | Minneapolis, MN | Mar 2011 – Apr 2017

Progressed from software developer to technical lead for enterprise retail, pharmacy, and infrastructure automation.

- **Engineering & delivery.** Led cross-shore teams of 10+ delivering applications and automation in C#, .NET, JavaScript, and SQL Server; owned implementation, testing, releases, and support.
- **Healthcare applications.** Supported Target RedRx pharmacy workflows for prescription fulfillment and claims adjudication, including database development, production support, and performance troubleshooting.
- **Automation & reliability.** Developed infrastructure provisioning and monitoring automation, reducing manual effort by 60%; supported pharmacy fulfillment and claims-adjudication platforms.

TECHNICAL EXPERTISE

AI & Agentic Systems	LLMs, Generative AI, RAG, LangChain, Semantic Kernel, Google ADK, MCP, Dialogflow CX
Languages & Frameworks	Python, Java, C#, JavaScript, Bash, J2EE / Java EE, ASP.NET, Node.js
Cloud & Infrastructure	Azure, GCP, Docker, Kubernetes, Red Hat Enterprise Linux, CentOS
DevOps & Reliability	GitHub Actions, GitOps, Jenkins, JFrog Artifactory, CI/CD, SRE, performance testing
Observability	Datadog, Arize, Dynatrace, Grafana, Splunk, Fluentd
Databases & App Servers	PostgreSQL, IBM Db2, SQL Server, Oracle, Redis, Microsoft IIS, IBM WebSphere, Apache Tomcat

EDUCATION & PROFESSIONAL DEVELOPMENT

Boston University — M.S., Computer Science & Artificial Intelligence (in progress; expected Jan 2028)
Johns Hopkins University — Certificate, AI in Healthcare (Aug 2026)
The University of Texas at Austin — Post Graduate Program, Generative AI (2025–2026)
Karunya Institute of Technology and Sciences — B.E., Computer Science & Engineering (2010)